Network Services

Name: Hana
Reg.No: SAB01-4608
System Administration & Infrastructure
ITSimplera Solutions
Instructor: Tahir (Senior System Administrator)

Table Of Contents:

Introduction

DNS (Domain Name System) and DHCP (Dynamic Host Configuration Protocol) are essential for smooth communication, centralized name resolution, and automated IP address management in enterprise networks. Prior to deploying Active Directory Domain Services (AD DS), each system administrator needs to comprehend how these fundamental network services function and communicate in a Windows Server setting.

Objective

The objective of this work is to configure, validate, debug, and implement DNS and DHCP services on Windows Server 2019 while preparing the infrastructure for the eventual deployment of Active Directory.

Theory Topics

Windows Server Network Services

- Hostname & Naming Standards: short/safe characters/avoid hardcoding names (use aliases)
- Network Interface Configuration: the setup of an IP address, subnet mask, and gateway to connect a device to a network.

  ### *Domain Name System (DNS)*
- DNS Resolution Process: the process by which DNS converts a domain name into an IP address.

- DNS Hierarchy:
    1. Root Level: managed by root DNS servers
    2. Top Level Domain: managed by TLD name servers (.com, .gov, .edu)
    3. Second Level Domain: organization registered name
    4. Subdomains: additional organizational structuring of a website (e.g .blog)
    5. Hosts: particular device
- Forward & Reverse Lookup Zones: A Forward Lookup Zone translates domain names into IP addresses, while a Reverse Lookup Zone does the opposite: it maps IP addresses back to domain names.
- DNS Records: distributed database filled with different types of records, each serving a unique purpose in this process
  A (address): map domain names to their corresponding IPv4 addresses.
  AAAA (quad-A): map domain names to their IPv6 addresses
  PTR (pointer): maps an IP address to a domain name (opposite of A)
  CNAME(canonical names): maps an alias or subdomain to another domain name (points one domain to another)
  MX (mail exchange): specify the mail servers responsible for receiving e-mail on behalf of a domain.
  NS (name server): specify the authoritative name servers that hold the DNS records for a particular domain.
- DNS Cache: saves past domain name lookups
- DNS forwarders: sends unresolved queries to a specific outer server.
- DNS Hijacking/ DNS redirection, is a type of DNS attack in which DNS queries are incorrectly resolved in order to unexpectedly

redirect users to malicious sites (Types: Local/Router/MITM/Rogue)

Dynamic Host Configuration Protocol (DHCP)

- Scope: configured range of IP addresses that a DHCP server is authorized to lease to clients on a given subnet
- Lease: the length of time that the client can use the IP address it has been assigned
- Reservation: links a device's unique physical identifier (MAC address) to a specific, permanent IP address.
- Exclusion Range: a configuration that tells a DHCP server not to dynamically assign certain IP addresses that fall within a defined scope
- Options*:* extra network configuration settings that a server sends to a device along with its IP address (e.g. 3= default router, 6= dns server address)
- Relay Agent: a router or Layer 3 switch configured to forward DHCP broadcast messages from clients on one subnet to a DHCP server located on a different subnet.
- APIPA Addressing: utomatic Private IP Addressing (APIPA) allows devices to self-assign dynamic IP addresses if they cannot do so through a DHCP server. The possible range of APIPA addresses is limited between 169.254.0.1 and 169.254.255.254, giving it a subnet mask of 255.255.0.0 (which can, alternatively, be displayed as “/16”).

Preparing for Active Directory

- Why DNS is Required for AD : Active Directory DNS is used to locate domain controllers and critical services (LDAP, Kerberos, and the Global Catalog) via SRV and host records. If DNS is missing or misconfigured, common outcomes include failed logons, Group Policy errors, and domain controller replication issues.

- Why DHCP Simplifies Enterprise Networks : It ensures that IP management is centralized and automated. It also reduces the risk of conflicts by ensuring that no two devices are assigned the same address. This is crucial in maintaining the integrity and stability of the network.
- Workgroup vs Domain
  domain: a centralized administration and all devices can be managed from a centralized device (remote login)
  workgroup: distributed administration wherein each user can manage his machine independently
- Active Directory Domain Services : Active Directory Domain Services (AD DS) are the core functions in Active Directory that manage users and computers and allow sysadmins to organize the data into logical hierarchies.
  Domain Services: Stores data and manages communications between the users and the DC. This is the primary functionality of AD DS.

Software Requirements

Software: VM workstation
OS Media: Windows Server 2019
Environment: Two Windows 10 VMs
Access Level: Administrator Account

Implementation

Windows Server Preparation

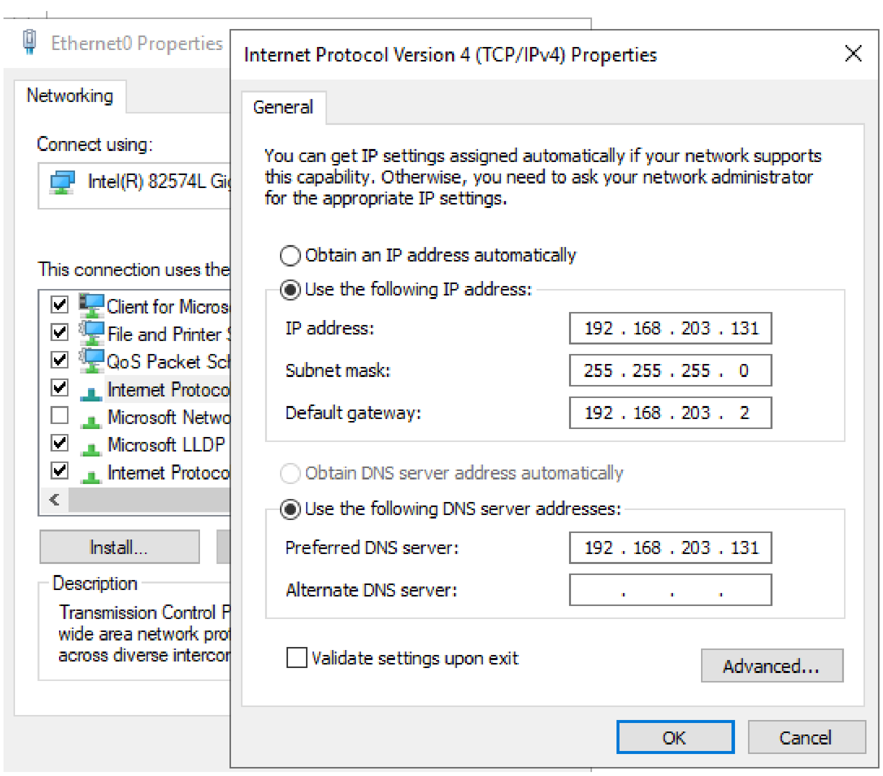
(configuring basic network settings and verifying connectivity)

Renaming the server to SRV-DC01

Configuring a Static IPv4 Address, Subnet Mask, Default Gateway, and Preferred DNS Server

start menu/power shell -> ncpa.cpl

Properties -> Internet Protocol Version 4 (TCP/IPv4) -> Use the following IP address



```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : SRV-DC01
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-10-4D-32
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::204c:8130:88bf:b759%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.203.131(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.203.2
   DHCPv6 IAID . . . . . . . . . . . : 83889193
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-E8-49-04-00-0C-29-10-4D-32
   DNS Servers . . . . . . . . . . . : 192.168.203.131
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Testing internal gateway

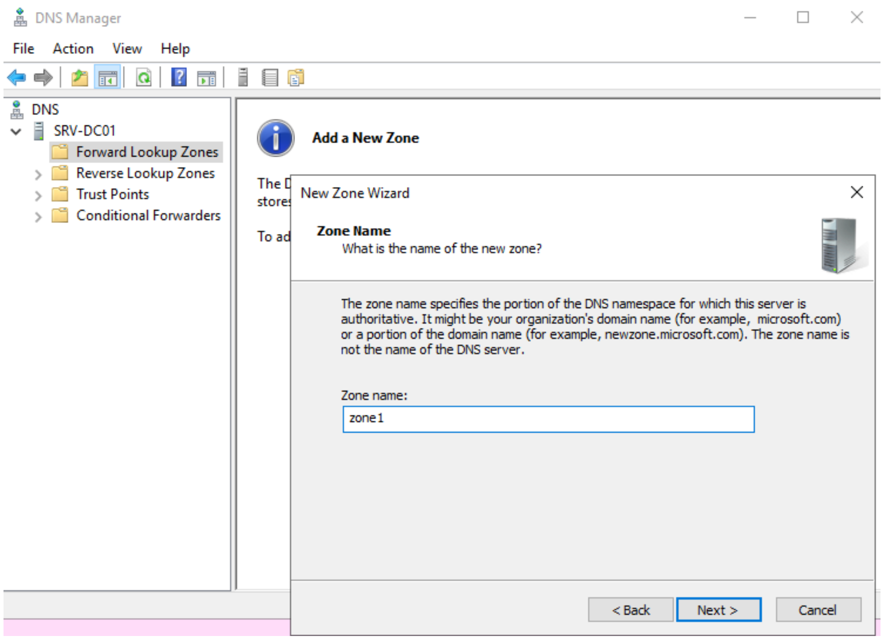
```
C:\Users\Administrator>ping 192.168.203.2

Pinging 192.168.203.2 with 32 bytes of data:
Reply from 192.168.203.2: bytes=32 time<1ms TTL=128
Reply from 192.168.203.2: bytes=32 time<1ms TTL=128
Reply from 192.168.203.2: bytes=32 time<1ms TTL=128
Reply from 192.168.203.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.203.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

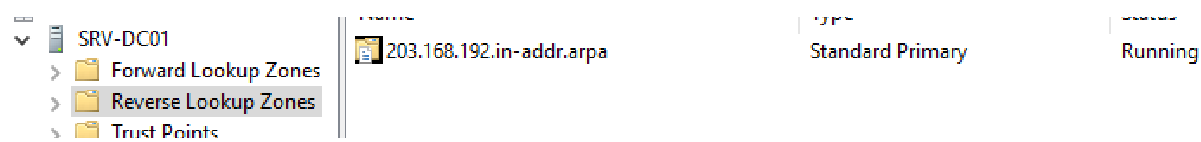
Testing external gateway (google)

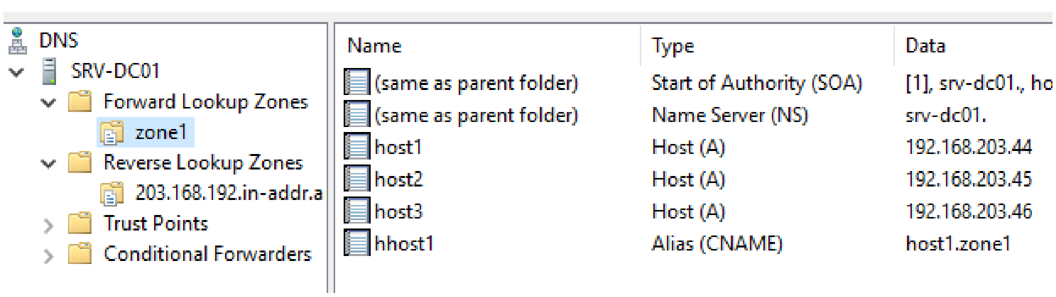
```
C:\Users\Administrator>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=55ms TTL=128
Reply from 8.8.8.8: bytes=32 time=52ms TTL=128
Reply from 8.8.8.8: bytes=32 time=57ms TTL=128
Reply from 8.8.8.8: bytes=32 time=56ms TTL=128

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 52ms, Maximum = 57ms, Average = 55ms
```

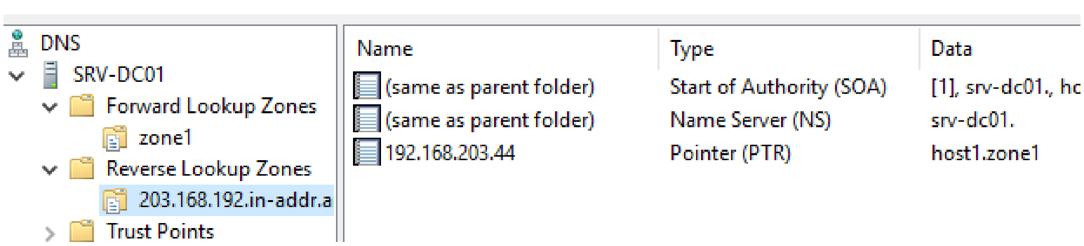
DNS Server Deployment & Configuration

Tools -> DNS -> create forward/reverse zones
inside a forward zone -> right click -> new host
inside a reverse zone -> right click -> new ptr

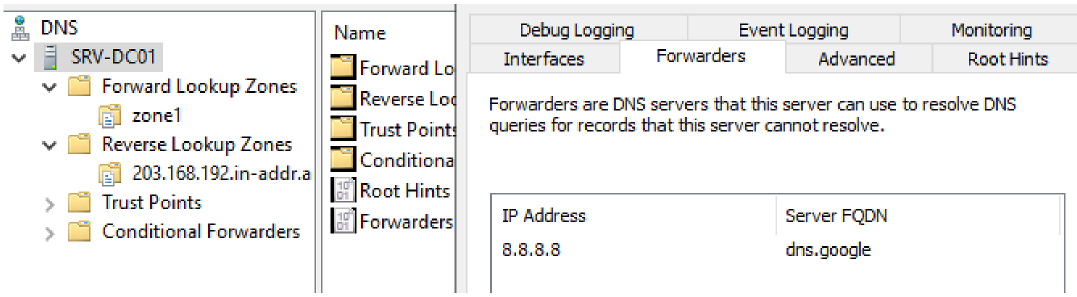








DNS Forwarders: DNS Manager -> right-click on Server Name -> Properties -> Forwarders tab -> Edit -> enter public DNS server addresses to allow resolution of external internet addresses



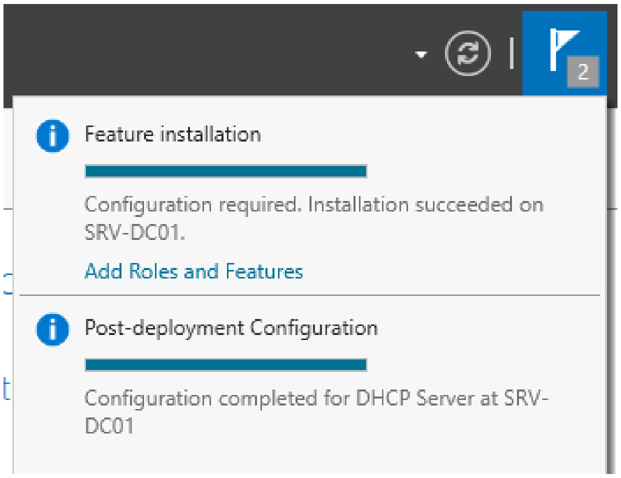
Verification:

```
C:\Users\Administrator>nslookup
Default Server:  one.one.one.one
Address:  1.1.1.1

> server 127.0.0.1
Default Server:  [127.0.0.1]
Address:  127.0.0.1

> host1.zone1
Server:  [127.0.0.1]
Address:  127.0.0.1

Name:    host1.zone1
Address:  192.168.203.44

> host2.zone1
Server:  [127.0.0.1]
Address:  127.0.0.1

Name:    host2.zone1
Address:  192.168.203.45

> hhost1.zone1
Server:  [127.0.0.1]
Address:  127.0.0.1

Name:    host1.zone1
Address:  192.168.203.44
Aliases:  hhost1.zone1
```

(test names & IP addresses -> changed later for correct ones)

DHCP Server Deployment & Configuration

Install DHCP role



Tools -> DHCP -> IPv4 -> right click -> new scope

<u>Name:</u> SCONE

<u>length/prefix</u> : /24

<u>Number of usable IP address</u> : 256 (usable 254)

<u>Subnet mask</u>: 255.255.255.0

<u>Start IP address</u>: 192.168.203.100

<u>End IP address</u>: 192.168.203.160

<u>Exclusion range:</u> 192.168.203.120 -> 192.168.203.140
(addresses or range of addresses that are not distributed by the server)

<u>Delay </u>: 10 ms
(time duration by which the server will delay the transmission of a DHCPOFFER message)

<u>Lease Duration:</u> 10 days : 10 hours : 30 min
(how long a client can use an IP address from this scope)

Default Gateway: 192. 168.203.2

Tools -> DHCP -> IPv4 -> Scope[start IP].name -> reservation -> right click -> new reservation

Reservation "RESV-1": 192.168.203.128 (VM1)
(ensures that a DHCP client is always assigned the same IP address)

verification
ipconfig /all

```
Connection-specific DNS Suffix  . : localdomain
Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . . . . . : 00-0C-29-5E-AA-D2
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8b92:a47c:89d:42aa%6(Preferred)
IPv4 Address. . . . . . . . . . . : 192.168.203.128(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Lease Obtained. . . . . . . . . . : 22 July 2026 19:59:27
Lease Expires . . . . . . . . . . : 22 July 2026 20:37:52
Default Gateway . . . . . . . . . : 192.168.203.2
DHCP Server . . . . . . . . . . . : 192.168.203.254
DHCPv6 IAID . . . . . . . . . . . : 100666409
DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-D8-8B-E4-00-0C-29-5E-AA-D2
DNS Servers . . . . . . . . . . . : 192.168.203.2
Primary WINS Server . . . . . . . : 192.168.203.2
NetBIOS over Tcpip. . . . . . . . : Enabled
```

Client Connectivity Verification

Release and Renew the IP Address

[ipconfig /release]

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::8b92:a47c:89d:42aa%6
Default Gateway . . . . . . . . . :
```

[ipconfig /renew]

```
Connection-specific DNS Suffix  . :
Link-local IPv6 Address . . . . . : fe80::8b92:a47c:89d:42aa%6
IPv4 Address. . . . . . . . . . . : 192.168.203.128
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . : 192.168.203.2
```

Display complete network information [ipconfig /all]

```
Windows IP Configuration

   Host Name . . . . . . . . . . . . : SRV-DC01
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-10-4D-32
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::204c:8130:88bf:b759%10(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.203.131(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.203.2
   DHCPv6 IAID . . . . . . . . . . . : 83889193
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-E8-49-04-00-0C-29-10-4D-32
   DNS Servers . . . . . . . . . . . : 192.168.203.131
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Verify DNS Name Resolution

```
 DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-D8
 DNS Servers . . . . . . . . . . . : 192.168.203.131
```

```
C:\Users\Administrator>nslookup
Default Server:  VM1.desktop-ij4iosm
Address:  192.168.203.131
```

```
C:\Users\User1>nslookup
Default Server:  VM1.desktop-ij4iosm
Address:  192.168.203.131
```

Verify communication with the Server [ping ip]

```
C:\Users\User1>ping 192.168.203.131

Pinging 192.168.203.131 with 32 bytes of data:
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128
Reply from 192.168.203.131: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.203.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Verify automatic DHCP address assignment

```
DHCP Enabled. . . . . . . . . . . : Yes
```

Challenges faced

1. Can’t create new reservation after creating DHCP

   Solution: right click on Scope [192.168.203.0] SCONE -> activate

2. DNS name resolution verification failed (connected to the wrong IP address)

```
C:\Users\User1>hostname
DESKTOP-IJ4IOSM

C:\Users\User1>nslookup DESKTOP-IJ4IOSM
Server:  one.one.one.one
Address:  1.1.1.1

*** one.one.one.one can't find DESKTOP-IJ4IOSM: Non-existent domain

C:\Users\User1>
```

```
DNS Servers . . . . . . . . . . . : 1.1.1.1
                                    1.0.0.1
```

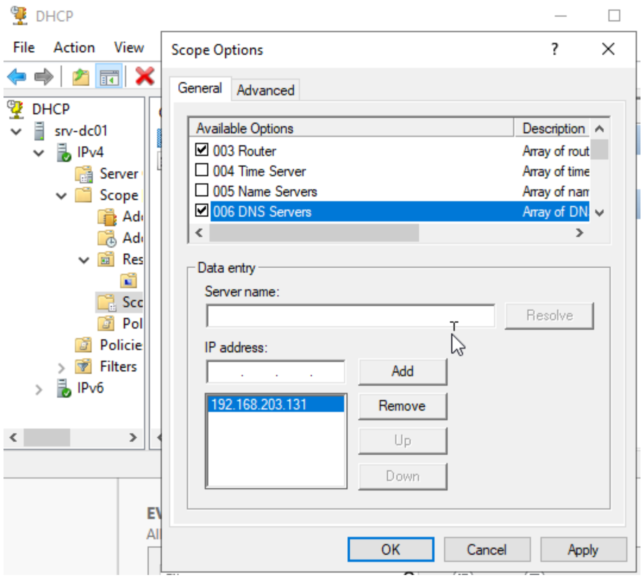
   using Cloudflare instead of your Windows DNS server.

   Attempted Solution: restart the client VM

```
C:\Users\User1>nslookup DESKTOP-IJ4IOSM
Server:  UnKnown
Address:  192.168.203.2

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to UnKnown timed-out
```

   After restarting, the DNS server is now 192.168.203.2 : the client is asking the **gateway** not the Windows Server for DNS.

Attempted Solution: dhcp -> socpe & server options -> set 006 DNS Server to the Windows server IP, and remove other addresses

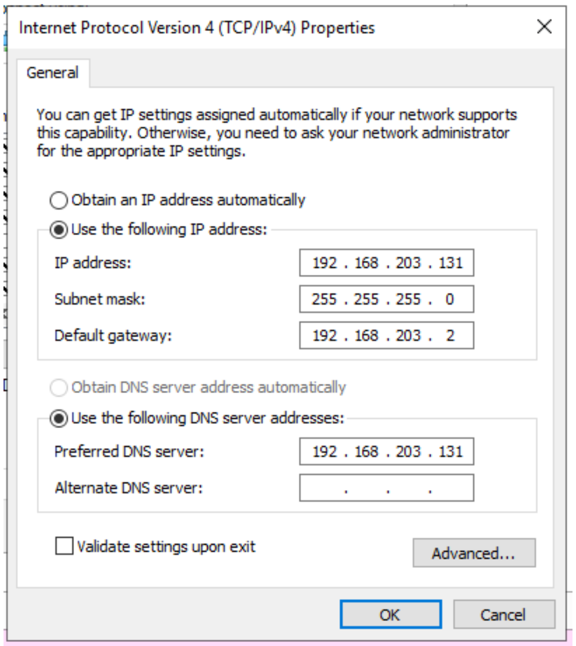


only worked temporarily, and DNS server doesn't have a DNs record for client's hostname

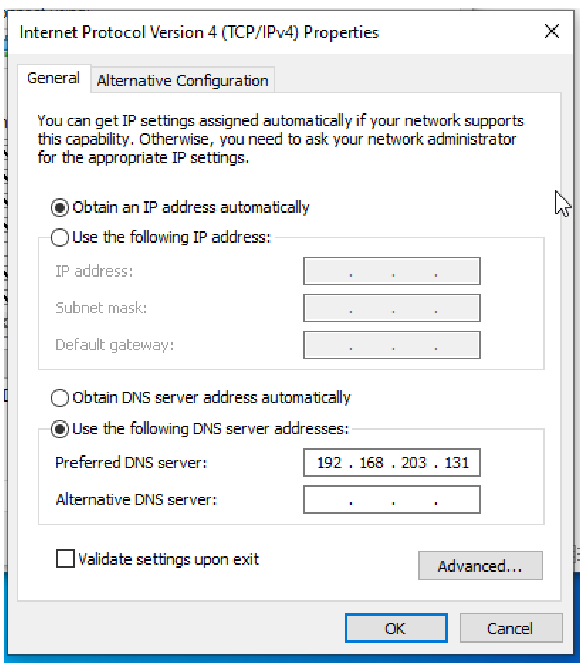
The client (DESKTOP-IJ4IOSM) grabbed its IP address via DHCP, but it hasn't successfully registered its A (Host) record into your Windows DNS Server zone yet.

The "Server failed" error happens because your Windows DNS server has no Host (A) record for DESKTOP-IJ4IOSM inside your zone, or the DNS suffix isn't matching up automatically.

Solution: match dns zone and VM names + set prefered dns server IP in client machine (same as dns server IP in server/ same server IP address)



Server



Client

3. Incorrect DHCP Scope

Solution: match DHCP and client machines subnet mask

Learning outcomes

- Network configuration
- Roles/features installation
- Static IP vs DHCP (scope/lease/architecture/exclusion)
- DNS resolution

Conclusion

This project demonstrated that correct configuration of DNS, DHCP, and IP addressing is essential for building a stable network infrastructure and preparing the environment for Active Directory Domain Services (AD DS).

References

- https://www.reddit.com/r/sysadmin/comments/13c7tom/server_naming_standards/
- https://learn.microsoft.com/en-us/troubleshoot/windows-server/active-directory/naming-conventions-for-computer-domain-site-ou
- https://serverfault.com/questions/17274/naming-conventions
- https://cycle.io/learn/dns-resolution-process
- https://hostnoc.com/dns-hierarchy-everything-you-need-to-know/?srsltid=AfmBOooMgVWARnGx4Q0q2k-wk2mYwVARP5GT8WcxhiRP9Ws1auw06yaA

- https://qrator.net/library/learning-center/DNS/What-Are-DNS-Records/?utm_referrer=https%3A%2F%2Fwww.google.com%2F#dns-ns-records
- https://www.imperva.com/learn/application-security/dns-hijacking-redirection/
- https://www.exam-labs.com/blog/the-backbone-of-network-connectivity-how-dhcp-simplifies-enterprise-management
- https://www.geeksforgeeks.org/computer-networks/difference-between-domain-and-workgroup/
- https://www.varonis.com/blog/active-directory-domain-services